

Edward Phillips

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I am a DPhil candidate in Engineering Science at the University of Oxford, working in the Computational Health Informatics Lab under Professor David Clifton. My research develops reliable intelligent systems for clinical decision-making, including uncertainty quantification in large language models, guideline-grounded treatment planning agents, and early disease diagnosis from large-scale hospital records. I previously spent three years in industry as a data scientist and systems engineer, working in generative AI, medical device development, and biomarker research. Most recently, I completed a research internship at Spotify, developing efficient and uncertainty-aware LLM ranking systems. I hold a Master's degree in Information and Computer Engineering from the University of Cambridge, achieving First Class with Distinction.

University Education

University of Oxford, Computational Health Informatics Lab

Oxford, UK

DPhil, Engineering Science, supervised by Professor David Clifton

October 2024 - Present

- The aims of the lab are to make fundamental contributions to the theory of AI in medicine while also working with clinical collaborators to produce systems that help patients. My research develops AI tools to support clinical decision-making, with an emphasis on explainable and uncertainty-propagating methods.
- I have published methods to quantify uncertainty in large language models, with applications in hallucination detection and mitigation.
- I created an agent-based framework for treatment planning in oncology, with an upcoming pilot evaluation in collaboration with NHS and industry partners (TrustedMDT).
- I am currently developing foundation models for electronic health record data as part of CODETECT, a multi-site NHS study identifying hospitalized patients with undiagnosed chronic conditions.

University of Cambridge

Cambridge, UK

MEng Information and Computer Engineering

September 2016 - June 2020

- I achieved a First in every year of examinations, and was awarded a senior scholarship each year for academic achievement. In my second year I was awarded the Bill Ray Prize for special aptitude in electrical subjects.
- I was awarded a distinction for my Master's project in computational neuroscience, in which I used TensorFlow to train biologically-interpretable models which could replicate the input-output mapping of real neurons.
- I specialized in Information and Computer Engineering. Particular areas of focus were deep learning, statistical signal analysis, and mathematical methods. The specific modules I studied included software design, deep learning, computer vision, advanced control theory, medical imaging, molecular biology, and neuroscience.

Publications

Conferences & Journals

Semantic Self-Distillation for Language Model Uncertainty

E. Phillips, S. Wu, F. K. Gustafsson, B. Gao, D. A. Clifton. *Uncertainty in Artificial Intelligence (UAI)* 2026.

Geometric Uncertainty for Detecting and Correcting Hallucinations in LLMs

E. Phillips, S. Wu, S. Molaei, D. Belgrave, A. Thakur, D. A. Clifton. *Transactions on Machine Learning Research (TMLR)* 2026.

BAS: A Decision-Theoretic Approach to Evaluating Large Language Model Confidence

S. Wu, F. K. Gustafsson, E. Phillips, B. Gao, A. Thakur, D. A. Clifton. *Conference on Language Modeling (COLM)* 2026.

Preprints & Under Review

When Text and Numbers Disagree: Evidence Arbitration in Large Language Models

M. Carletti, E. Phillips, F. K. Gustafsson, P. Palo, L. Clifton, D. Belgrave, X. Gu, D. A. Clifton. arxiv.org/abs/2608.20116

Entropy Alone is Insufficient for Safe Selective Prediction in LLMs

E. Phillips, F. K. Gustafsson, S. Wu, A. Thakur, D. A. Clifton. arxiv.org/abs/2603.21172

Foundation Models for Biosignals: A Survey

X. Gu, Y. Shu, J. Han, Y. Liu, Z. Liu, J. Anibal, V. Sangha, **E. Phillips**, B. Segal, Y. Liu, H. Yuan, F. Liu, K. Branson, P. Schwab, D. Belgrave, L. Clifton, D. Spathis, V. Lampos, A. A. Faisal, D. A. Clifton.
techrxiv.org/doi/full/10.36227/techrxiv.175606236.62808131

Employment History

Spotify

London, UK

Research Scientist Intern

June - September 2026

- I explored the efficient use of LLMs in ranking systems, including uncertainty-based decision policies and reliable integration with behavioural priors.
- Using Bayesian routing policies, we developed LLM ranking systems which cut latency by up to 70% without performance degradation.

Norstellia - pharmaceutical solutions

London, UK

Data Scientist

November 2023 - September 2024

- I developed generative AI applications for customers across the pharmaceutical industry, including a retrieval-augmented generation framework to facilitate interrogation of Norstellia's forecasting and market analysis datasets.
- I used the suite of tools in the AWS ecosystem to develop, deploy and manage data science services. I am familiar with AWS cloud services, REST/Websocket APIs, Docker and terraform, among other tools.
- I integrated new predictive models into Norstellia's suite of financial forecasting products, using interpretable machine learning approaches and deployment and management tools such as Apache Airflow and Kubernetes.

Owlstone Medical - medtech scale-up developing a breathalyzer for disease

Cambridge, UK

Data Scientist

June 2022 - April 2023

- I worked across multiple projects applying machine learning and data science to support R&D, primarily working with time-series sensor data to predict concentrations of on-breath gases, enabling diagnosis of digestive diseases via a handheld device.
- I created a cost-efficient digital pathology pipeline to quantify protein expression in high-resolution tissue images, delivering same-day results at less than 20% of the cost of commercial alternatives. The pipeline identified promising candidate biomarkers relating to early identification of cancers.
- I applied transformer-based protein foundation models and graph neural networks to predict enzyme-substrate affinities, which accelerated selection of suitable compounds for use in Owlstone's diagnostics platform.

Owlstone Medical

Cambridge, UK

Systems Engineer

April 2021 - June 2022

- I led the engineering work to design, develop, and test a mouse breath sampling device for use in biomarker discovery. I wrote a Python-based GUI for user interaction, I developed a dynamic control system to regulate air flows, and managed a multi-device electronics configuration. The system was used in an animal study to derisk the company's flagship method for breath-based detection of lung cancer.

Organox - organ preservation for transplantation

Oxford, UK

Engineering Intern

July - August 2018

- I researched how to scale down the company's organ-preserving perfusion device to work for small organs such as kidneys. I developed a test circuit and wrote control firmware in C to deliver controlled oxygen flow rates in an energy-efficient manner.
- Using a form of oxygen storage I was able to achieve an 85% reduction in energy consumption, and demonstrated how further savings could be achieved. Organox continued to develop a kidney perfusion device with a successful human clinical trial completed in 2024.

Livingbridge Private Equity

London, UK

Junior Analyst

July - September 2017

- I created relational databases for sensitive company data, and wrote SQL and Excel VBA for analytics tasks.

Teaching

University of Oxford, Engineering Science Undergraduate Tutorials

Oxford, UK

Machine Learning, Software Engineering

2025 - Present

- I teach two courses to third-year Engineering Science undergraduates. The machine learning course covers topics including graphical models, approximate inference, and neural networks. The software engineering course covers C++ design principles and algorithmic development.

University of Oxford, Lectures and Labs

Oxford, UK

Centre for Doctoral Training in Healthcare Data Science, MSc Applied Digital Health

2024 - Present

- Together with colleagues from my research lab, I teach modules in CDT and MSc courses covering Python, Machine Learning, and Artificial Intelligence, with a focus on healthcare applications.

Skills

Programming Languages/Tools: Python, SQL, AWS, Azure, Git

Libraries: PyTorch, TensorFlow, Transformers, Numpy, pandas, scikit-learn

Languages: English (native), French (intermediate), Spanish (intermediate)

Academic Service & Outreach

ACL Rolling Review (ARR)

Reviewer

2026

BMJ Digital Health & AI

Reviewer

2025 – Present

ICLR Workshop: Agentic AI in the Wild

Reviewer

2026

STIMULUS Community Service Programme

Cambridge, UK

Maths Teaching Assistant

2018 – 2019

I provided differentiated support to secondary school maths students, working individually with those needing additional help and stretching more confident students with extension problems.